StoreX

Bottles & Carboys

REAGENT BOTTLES

PlastX

www.plastx.in

The PlastX Factor

Leakproof Guarantee

PlastX bottles and closures are designed with a strong and semi-buttress thread design. We offer a leakproof guarantee as we manufacture and test both components routinely as part of our quality inspection process. PlastX closures come with no liner that wears or corrodes, or causes contamination.

High Quality Resin

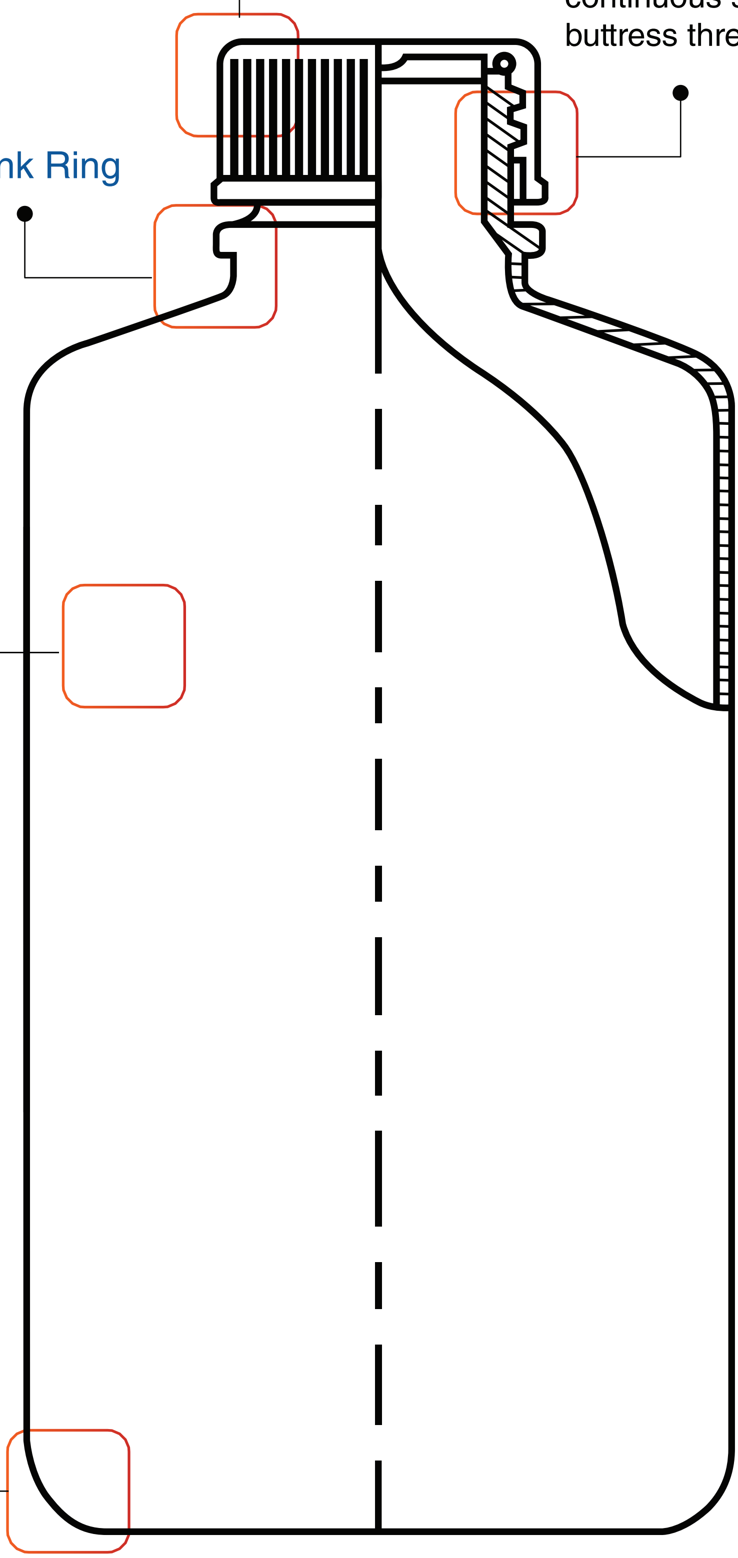
The PlastX range of bottles & carboys are made from the highest quality resin that meet medical grade standards. Our resins are selected to minimize additives and reduce potential leachables.
We don't use plasticizers or fillers, thus creating a safe storage space for your valuable samples.

Shatter-proof Safe

Plastics contain lower concentrations of trace elements extractables than glass and are less likely to break. PlastX containers amplify this advantage, creating rugged containers that assure you the protection you expect for your valuable work.

Discover reliable Bottles

The PlastX range of Bottles are the classic standard storage containers. They offer the ultimate solution in containment & protection. Available in 3 different resins, they are ensured to be leakproof, strong and durable.



Threads

on both bottles & closures are smooth, continuous semi buttress threads.

Closure

Shrink Ring

Uniform Thickness

Bottom

curved inner corners for superior cleaning

Symbol Glossary

USP Class VI certified medical grade resin

121°C 15 psi — Products listed as Autoclavable can be autoclaved at 121°C, 15 psi for 20 minutes

Narrow Mouth Bottles

Narrow-Mouth Bottle - PP

PlastX Narrow-Mouth Bottles are ideal for lab applications that require excellent chemical resistance. Autoclavable with and without contents.

- Caps provide the ultimate leakproof protection.
- Provides excellent chemical resistance for a variety of lab applications.
- Translucence provides good clarity for viewing contents.
- 2, 4 and 8 L Narrow-Mouth Bottles feature a built-in shoulder loop for attaching identification tag.
- Autoclavable and Leakproof.

Medical Grade Resin and Packing USP Class VI

121°C 15 psi

Code No.	Description	Pack Size
100101	4ML	72
100102	8ML	72
100103	15ML	72
100104	30ML	72
100105	60ML	72
100106	125ML	72
100107	250ML	72
100108	500ML	48
100109	1000ML	24

Narrow-Mouth Bottle - LDPE

PlastX Narrow-Mouth Bottles are ideal for lab applications that require excellent chemical resistance. Medical grade confirming to USP Class VI.

- Excellent chemical resistance to most acids, bases and alcohols make this a versatile bottle.
- Can be used in freezer to -100°C (-148°F) for long-term storage.
- Translucence allows easy viewing of liquid levels.
- Caps provide the ultimate leakproof protection.

Medical Grade Resin and Packing USP Class VI

Code No.	Description	Pack Size
100702	15ML	72
100703	30ML	72
100704	60ML	72
100705	125ML	72
100706	250ML	72
100707	500ML	48
100708	1000ML	24

Narrow-Mouth Bottle - HDPE

PlastX Narrow Mouth Bottles are suitable for general purpose laboratory applications.

- Caps ensure leakproof performance.
- Reliable durability.
- Can be used for packing and shipping purpose.

Medical Grade Resin and Packing USP Class VI

Code No.	Description	Pack Size
100301	4ML	72
100302	8ML	72
100303	15ML	72
100304	30ML	72
100305	60ML	72
100306	125ML	72
100307	250ML	72
100308	500ML	48
100309	1000ML	24

Amber Narrow-Mouth Bottle - HDPE

PlastX Amber Narrow Mouth Rectangular Bottles help protect light sensitive liquids or dry materials from exposure.

- Amber HDPE reduces UV light transmissions to protect light sensitive liquids.
- Excellent chemical resistance to most acids, bases and alcohols helps these containers protect your valuable samples.
- Caps provide the ultimate in leakproof protection.

Medical Grade Resin and Packing USP Class VI

Code No.	Description	Pack Size
100501	4ML	72
100502	8ML	72
100503	15ML	72
100504	30ML	72
100505	60ML	72
100506	125ML	72
100507	250ML	72
100508	500ML	48
100509	1000ML	24

Wide Mouth Bottles

Wide-Mouth Bottle - PP

PlastX Wide-Mouth Economy Bottles are suitable for general purpose laboratory applications. Wide mouth is easy to fill.

- Caps ensure leakproof performance.
- Autoclavable.
- Leakproof.

121°C
15 psi

Code No.	Description	Pack Size
100201	30ML	72
100202	60ML	72
100203	125ML	72
100204	250ML	72
100205	500ML	48
100206	1000ML	24
100207	2000ML	6

Wide-Mouth Bottle - LDPE

PlastX Wide-Mouth Bottles are suitable for general purpose laboratory applications. Wide mouth is easy to fill.

- Caps ensure leakproof performance.
- Leakproof.

Code No.	Description	Pack Size
100801	30ML	72
100802	60ML	72
100803	125ML	72
100804	250ML	72
100805	500ML	48
100806	1000ML	24
100807	2000ML	6

Wide-Mouth Bottle - HDPE

PlastX Wide-Mouth Bottles are suitable for general purpose laboratory applications.

- Caps ensure leakproof performance.
- Reliable durability.
- Can be used for packing and shipping purpose.
- Leakproof.

Code No.	Description	Pack Size
100401	30ML	72
100402	60ML	72
100403	125ML	72
100404	250ML	72
100405	500ML	48
100406	1000ML	24
100407	2000ML	6

Amber Wide-Mouth Bottle - HDPE

PlastX Amber Wide Mouth Rectangular Bottles help protect light sensitive liquids or dry materials from exposure.

- Amber HDPE reduces UV light transmissions to protect light sensitive liquids.
- Excellent chemical resistance to most acids, bases & alcohols helps these containers protect your valuable samples.
- Caps provide the ultimate in leakproof protection.

Code No.	Description	Pack Size
100601	30ML	72
100602	60ML	72
100603	125ML	72
100604	250ML	72
100605	500ML	48
100606	1000ML	24
100607	2000ML	6

Bulk Bottle Packaging

PlastX provides bulk bottle packaging solutions designed for high-volume laboratory and industrial workflows. These packaging formats support sectors such as diagnostics, specialty chemicals, etc., where efficient handling of large quantities of lab containers is essential.

Our bulk packing approach focuses on safe transport, organized storage, and operational efficiency. By optimizing the way bottles are packed and delivered, PlastX helps laboratories reduce handling time, maintain product integrity, and simplify inventory management.

Through StoreX, PlastX's dedicated storage and logistics system, bulk packaging is engineered to support smooth workflows from warehouse storage to production environments. The system also helps minimize packaging waste and reduces the chances of damage during shipping.

Key Advantages

Reliable Protection During Transit

Each batch is carefully packed using protective bagging and sturdy transport cartons to safeguard bottles and closures from contamination, dust, and physical damage during shipment.

Efficient Handling for Large Volumes

Designed for laboratories and production facilities that require large quantities, this packaging format allows faster unpacking and easier material management.

StoreX™ Smart Storage System

PlastX integrates the StoreX storage concept to ensure bulk products remain well organized, easy to access, and protected throughout storage and distribution.

Flexible Packaging Options

Bulk configurations can be adapted to suit different operational requirements, helping facilities receive containers that are ready to integrate into their workflows.

Automation-Compatible Format

Packaging is optimized for environments using automated filling, capping, and dispensing systems, supporting smooth production processes.

Ordering Information

Code No.	Capacity	Desciption	Closure Size	Unit Pack
100101D	4ML	Narrow Mouth Round Bottle Polypropylene	13-415	2000
100102D	8ML	Narrow Mouth Round Bottle Polypropylene	20-415	2000
100103D	15ML	Narrow Mouth Round Bottle Polypropylene	20-415	2000
100104D	30ML	Narrow Mouth Round Bottle Polypropylene	20-415	1000
100105D	60ML	Narrow Mouth Round Bottle Polypropylene	20-415	1000
100106D	125ML	Narrow Mouth Round Bottle Polypropylene	24-415	500
100107D	250ML	Narrow Mouth Round Bottle Polypropylene	24-415	400
100108D	500ML	Narrow Mouth Round Bottle Polypropylene	28-415	200
100109D	1000ML	Narrow Mouth Round Bottle Polypropylene	38-430	100
100301D	4ML	Narrow Mouth Round Bottle HDPE Natural	13-415	2000
100302D	8ML	Narrow Mouth Round Bottle HDPE Natural	20-415	2000
100303D	15ML	Narrow Mouth Round Bottle HDPE Natural	20-415	2000
100304D	30ML	Narrow Mouth Round Bottle HDPE Natural	20-415	1000
100305D	60ML	Narrow Mouth Round Bottle HDPE Natural	20-415	1000
100306D	125ML	Narrow Mouth Round Bottle HDPE Natural	24-415	500
100307D	250ML	Narrow Mouth Round Bottle HDPE Natural	24-415	400
100308D	500ML	Narrow Mouth Round Bottle HDPE Natural	28-415	200
100309D	1000ML	Narrow Mouth Round Bottle HDPE Natural	38-430	100
100301WD	4ML	Narrow Mouth Round Bottle HDPE White	13-415	2000
100302WD	8ML	Narrow Mouth Round Bottle HDPE White	20-415	2000
100303WD	15ML	Narrow Mouth Round Bottle HDPE White	20-415	2000
100304WD	30ML	Narrow Mouth Round Bottle HDPE White	20-415	1000
100305WD	60ML	Narrow Mouth Round Bottle HDPE White	20-415	1000
100306WD	125ML	Narrow Mouth Round Bottle HDPE White	24-415	500
100307WD	250ML	Narrow Mouth Round Bottle HDPE White	24-415	400
100308WD	500ML	Narrow Mouth Round Bottle HDPE White	28-415	200
100309WD	1000ML	Narrow Mouth Round Bottle HDPE White	38-430	100
100501D	4ML	Narrow Mouth Round Bottle HDPE Amber	13-415	2000
100502D	8ML	Narrow Mouth Round Bottle HDPE Amber	20-415	2000
100503D	15ML	Narrow Mouth Round Bottle HDPE Amber	20-415	2000
100504D	30ML	Narrow Mouth Round Bottle HDPE Amber	20-415	1000
100505D	60ML	Narrow Mouth Round Bottle HDPE Amber	20-415	1000
100506D	125ML	Narrow Mouth Round Bottle HDPE Amber	24-415	500
100507D	250ML	Narrow Mouth Round Bottle HDPE Amber	24-415	400
100508D	500ML	Narrow Mouth Round Bottle HDPE Amber	28-415	200
100509D	1000ML	Narrow Mouth Round Bottle HDPE Amber	38-430	100
100201D	30ML	Wide Mouth Round Bottle Polypropylene	28-415	1000
100202D	60ML	Wide Mouth Round Bottle Polypropylene	28-415	1000
100203D	125ML	Wide Mouth Round Bottle Polypropylene	38-415	500
100204D	250ML	Wide Mouth Round Bottle Polypropylene	38-415	400
100205D	500ML	Wide Mouth Round Bottle Polypropylene	53-415	200
100206D	1000ML	Wide Mouth Round Bottle Polypropylene	63-430	100

Code No.	Capacity	Desciption	Closure Size	Unit Pack
100401D	30ML	Wide Mouth Round Bottle HDPE Natural	28-415	1000
100402D	60ML	Wide Mouth Round Bottle HDPE Natural	28-415	1000
100403D	125ML	Wide Mouth Round Bottle HDPE Natural	38-415	500
100404D	250ML	Wide Mouth Round Bottle HDPE Natural	38-415	400
100405D	500ML	Wide Mouth Round Bottle HDPE Natural	53-415	200
100406D	1000ML	Wide Mouth Round Bottle HDPE Natural	63-430	100
100601D	30ML	Wide Mouth Round Bottle HDPE Amber	28-415	1000
100602D	60ML	Wide Mouth Round Bottle HDPE Amber	28-415	1000
100603D	125ML	Wide Mouth Round Bottle HDPE Amber	38-415	500
100604D	250ML	Wide Mouth Round Bottle HDPE Amber	38-415	400
100605D	500ML	Wide Mouth Round Bottle HDPE Amber	53-415	200
100606D	1000ML	Wide Mouth Round Bottle HDPE Amber	63-430	100

Sterile Individually Wrapped

PlastX sterile bottles are manufactured using certified materials and are processed through sterilization to ensure dependability and purity.

Designed with very low trace elements, these bottles provide excellent protection against contamination and maintain the integrity of sensitive samples.

The bottles are leak-resistant, rigid, and chemically durable, making them suitable for handling a wide variety of laboratory reagents and solutions. Each bottle offers strong resistance to most acids, bases, and alcohols, ensuring reliable performance across laboratory, research, and industrial applications.

PlastX sterile bottles are produced from high-quality HDPE and are available in narrow-mouth configurations with PP closures. The design supports safe storage and easy dispensing while maintaining secure sealing during transport and use.

To preserve sterility during handling and transportation, every bottle is individually wrapped in sterile packaging and clearly labeled. **Packaging includes gamma indicator markings, providing visual confirmation that sterilization has been successfully completed.**

Select variants are also available as pyrogen-free, ensuring compatibility with applications that require minimal endotoxin levels.

Through PlastX's StoreX™ sterile packaging system, bottles remain protected throughout storage and distribution, allowing laboratories to maintain controlled and organized sterile inventory until the moment of use.

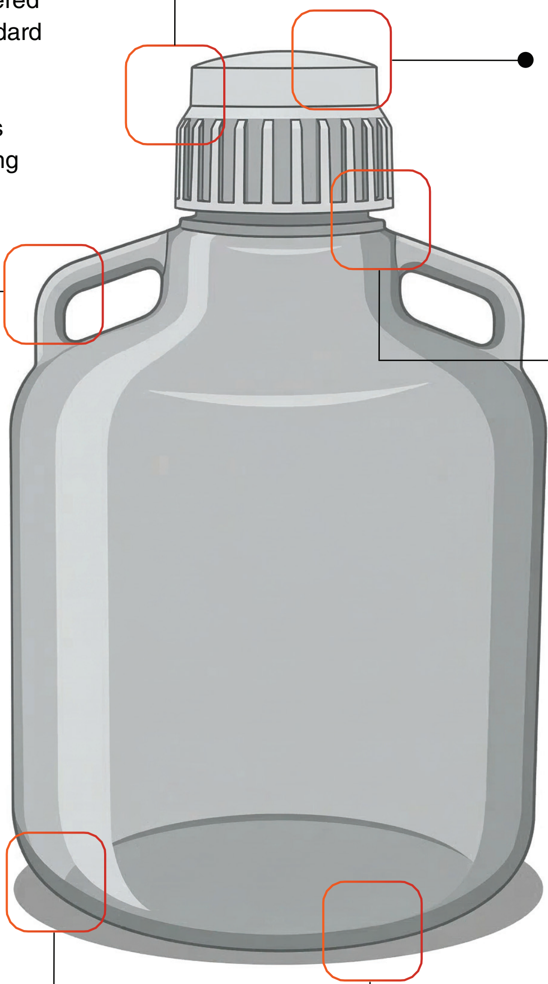
Ordering Information

Code No.	Capacity	Desciption	Closure Size	Unit Pack
100411	30ML	Wide Mouth Round Bottle HDPE Sterile	28-415	500
100412	60ML	Wide Mouth Round Bottle HDPE Sterile	28-415	250
100413	125ML	Wide Mouth Round Bottle HDPE Sterile	38-415	125
100414	250ML	Wide Mouth Round Bottle HDPE Sterile	38-415	60
100415	500ML	Wide Mouth Round Bottle HDPE Sterile	53-415	30
100416	1000ML	Wide Mouth Round Bottle HDPE Sterile	63-430	24
100611	30ML	Wide Mouth Round Bottle HDPE Amber Sterile	28-415	500
100612	60ML	Wide Mouth Round Bottle HDPE Amber Sterile	28-415	250
100613	125ML	Wide Mouth Round Bottle HDPE Amber Sterile	38-415	125
100614	250ML	Wide Mouth Round Bottle HDPE Amber Sterile	38-415	60
100615	500ML	Wide Mouth Round Bottle HDPE Amber Sterile	53-415	30
100616	1000ML	Wide Mouth Round Bottle HDPE Amber Sterile	63-430	24
100421	30ML	Wide Mouth Round Bottle HDPE Sterile	28-415	500
100422	60ML	Wide Mouth Round Bottle HDPE Sterile	28-415	250
100423	125ML	Wide Mouth Round Bottle HDPE Sterile	38-415	125
100424	250ML	Wide Mouth Round Bottle HDPE Sterile	38-415	60
100425	500ML	Wide Mouth Round Bottle HDPE Sterile	53-415	30
100426	1000ML	Wide Mouth Round Bottle HDPE Sterile	63-430	24
100621	30ML	Wide Mouth Round Bottle HDPE Amber Sterile	28-415	500
100622	60ML	Wide Mouth Round Bottle HDPE Amber Sterile	28-415	250
100623	125ML	Wide Mouth Round Bottle HDPE Amber Sterile	38-415	125
100624	250ML	Wide Mouth Round Bottle HDPE Amber Sterile	38-415	60
100625	500ML	Wide Mouth Round Bottle HDPE Amber Sterile	53-415	30
100626	1000ML	Wide Mouth Round Bottle HDPE Amber Sterile	63-430	24

Carboys

Discover what makes PlastX Carboys reliable

PlastX Carboys are designed to deliver durable liquid storage and dependable dispensing performance in laboratory and industrial environments. Built using high-quality materials and thoughtful engineering, these containers ensure safe handling, leak protection, and easy cleaning while supporting demanding workflows.



Closure

The closure is carefully engineered to be fully compatible with standard carboy and jerrican threading, creating a secure seal that minimizes leakage and ensures reliable liquid containment during storage and transport.

Seal Ring

A precision-moulded seal ring integrated within the closure fits tightly against the beveled edge of the bottle neck, helping maintain a consistent seal and preventing accidental leakage.

Built-in Shoulders

Ergonomically shaped integrated shoulders provide a comfortable grip and make lifting and carrying the carboy easier, especially when handling larger volumes.

Threads

Both the carboy neck and the closure feature continuous, straight-shouldered semi-buttress threads designed to improve grip and reduce the risk of thread damage or stripping during repeated use.

Bottom

The interior base is designed with smooth curved corners, reducing residue accumulation and allowing more efficient cleaning after use.

Stable Base

A reinforced base structure ensures stability during storage and use, and includes engraved identification details such as resin code and volume markings for easy reference.

Symbol Glossary

USP Class VI
Manufactured using medical-grade resin that meets USP Class VI standards.

Autoclavable
Suitable for sterilization at 121°C, 15 psi for 20 minutes.

Food Contact Safe
Materials are appropriate for applications involving contact with food products.

Sterile Indicator
Indicates products that have been sterilized through irradiation processes.

Carboys PP

Ideal for in-house autoclaving

Polypropylene Carboy

- Graduated polypropylene body with autoclavable construction, suitable for repeated sterilization cycles.
- Leak-resistant PP screw cap design ensures secure sealing and dependable performance during use.
- Wide shoulder handles allow comfortable carrying and controlled pouring of liquids.
- Wide-mouth configuration enables easier filling, cleaning, and dispensing operations.
- Heavy-duty versions available for vacuum applications or demanding laboratory conditions.
- Well-suited for storing and transporting large volumes of culture media, distilled water, and other laboratory solutions.

Recommended Applications

- Where autoclave sterilization is required
- Vacuum or demanding operating conditions (Heavy-Duty Carboy)
- Liquid storage and transfer applications (Narrow Mouth Carboy)
- Solid or powdered material storage (Wide Mouth Carboy)
- Where efficient storage space utilization is important (Rectangular shape)
- For uniform mixing or stirring processes (Round shape)

Ordering Information

Code No.	Capacity	Desciption	Closure Size (MM)	Pack Size
101801	5L	ASPIRATOR BOTTLE	53	1
101802	10L	ASPIRATOR BOTTLE	53	1
101803	20L	ASPIRATOR BOTTLE	53	1
101804		ASPIRATOR BOTTLE	53	1
101901	10L	CARBOYS	838	1
101902	20L	CARBOYS	838	1
102001	10L	CARBOYS WITH STOP CORK	838	1
102002	20L	CARBOYS WITH STOP CORK	838	1
102101	10L	WIDE MOUTH CARBOYS	100-415	1
102102	20L	WIDE MOUTH CARBOYS	100-415	1

Round Bottle

Nominal Capacity	Description	Approx. Brim Capacity ml	Material	Colour	Screw Closure	I.D. Neck mm	O.D. Bottle mm	Approx. Hgt without Closure mm
4 ml	Bottle NM	4.3	PP	Natural	13	8.3	15.8	38.8
	Bottle NM	4.3	HDPE	Natural	13	8.3	15.8	38.8
	Bottle NM	4.3	HDPE	Amber	13	8.3	15.8	38.8
8 ml	Bottle NM	11.1	PP	Natural	20	13.4	25	42.3
	Bottle NM	11.1	LDPE	Natural	20	13.4	24.8	42.3
	Bottle NM	11.1	HDPE	Natural	20	13.2	25	42.3
	Bottle NM	11.1	HDPE	Amber	20	13.2	25	42.3
15 ml	Bottle NM	17	PP	Natural	20	13.5	25	56
	Bottle NM	17	LDPE	Natural	20	13.3	24.8	55.8
	Bottle NM	17	HDPE	Natural	20	13.4	24.9	55.5
	Bottle NM	17	HDPE	Amber	20	13.4	24.9	55.5
30 ml	Bottle NM	35	PP	Natural	20	13.8	36.4	60
	Bottle NM	35	LDPE	Natural	20	13.8	36.4	60
	Bottle NM	35	HDPE	Natural	20	13.5	36.3	59.5
	Bottle NM	35	HDPE	Amber	20	13.5	36.3	59.5
	Bottle WM	34.5	PP	Natural	28	21.4	34.8	59.8
	Bottle WM	34.5	LDPE	Natural	28	21.4	34.7	59.8
	Bottle WM	34.5	HDPE	Natural	28	21.2	34.6	59.4
	Bottle WM	34.5	HDPE	Amber	28	21.2	34.6	59.4
60 ml	Bottle NM	69	PP	Natural	20	13.8	40	82.6
	Bottle NM	69	LDPE	Natural	20	13.8	39.9	82
	Bottle NM	69	HDPE	Natural	20	13.5	39.8	82
	Bottle NM	69	HDPE	Amber	20	13.5	39.8	82
	Bottle WM	67	PP	Natural	28	21.3	39.7	82.2
	Bottle WM	67	LDPE	Natural	28	21.3	39.6	82.1
	Bottle WM	67	HDPE	Natural	28	21.2	39.5	82
	Bottle WM	67	HDPE	Amber	28	21.2	39.5	82
125 ml	Bottle NM	140	PP	Natural	24	17.7	51.5	98.1
	Bottle NM	140	LDPE	Natural	24	17.7	51.5	98
	Bottle NM	140	HDPE	Natural	24	17.7	51.2	98.1
	Bottle NM	140	HDPE	Amber	24	17.7	51.2	98.1
	Bottle WM	139	PP	Natural	38	28.7	50.5	95.5
	Bottle WM	139	LDPE	Natural	38	28.6	50.4	95.4
	Bottle WM	139	HDPE	Natural	38	28.4	50.3	95.2
	Bottle WM	139	HDPE	Amber	38	28.4	50.3	95.2
250 ml	Bottle NM	270	PP	Natural	24	17.7	61.5	131.5
	Bottle NM	270	LDPE	Natural	24	17.7	61.3	131.2
	Bottle NM	270	HDPE	Natural	24	17.7	61	131
	Bottle NM	270	HDPE	Amber	24	17.7	61	131
	Bottle WM	269	PP	Natural	38	28.7	61.5	127.5
	Bottle WM	269	LDPE	Natural	38	28.6	61.2	127.4
	Bottle WM	269	HDPE	Natural	38	28.4	60	127.3
	Bottle WM	269	HDPE	Amber	38	28.4	60	127.3
500 ml	Bottle NM	540	PP	Natural	28	21.2	73.5	167.8
	Bottle NM	540	LDPE	Natural	28	21.2	73	167.7
	Bottle NM	540	HDPE	Natural	28	21.1	73	167.7
	Bottle NM	540	HDPE	Amber	28	21.1	73	167.7
	Bottle WM	530	PP	Natural	52.6	43.8	73.1	164.4
	Bottle WM	530	LDPE	Natural	52.6	43.8	72.5	164.3
	Bottle WM	530	HDPE	Natural	52.6	43.8	72.5	164.3
	Bottle WM	530	HDPE	Amber	52.6	43.8	72.5	164.3

Nominal Capacity	Description	Approx. Brim Capacity ml	Material	Colour	Screw Closure	I.D. Neck mm	O.D. Bottle mm	Approx. Hgt without Closure mm
1 litre	Bottle NM	1080	PP	Natural	38	27.8	93.8	213.5
	Bottle NM	1080	LDPE	Natural	38	27.8	93.3	213.2
	Bottle NM	1080	HDPE	Natural	38	27.8	93.2	213.5
	Bottle NM	1080	HDPE	Amber	38	27.8	93.2	213.5
	Bottle WM	1060	PP	Natural	63	53.3	92.5	192.8
	Bottle WM	1060	LDPE	Natural	63	52.2	92.1	192.5
	Bottle WM	1060	HDPE	Natural	63	51.9	92	192.5
	Bottle WM	1060	HDPE	Amber	63	51.9	92	192.5
2 litre	Bottle NM	2060	PP	Natural	38	27	120	244
	Bottle NM	2060	LDPE	Natural	38	27	119	242
	Bottle WM	2260	PP	Natural	100	90.5	119.5	236
	Bottle WM	2260	LDPE	Natural	100	90.2	119.2	236
	Bottle WM	2260	HDPE	Natural	100	89.5	119	236
	Bottle WM	2260	HDPE	Amber	100	89.5	118.5	236
	Heavy Duty Vacuum Bottle	2050	PP	Natural	53	38.4	120	255
4 litre	Bottle NM	4315	PP	Natural	38	27	153	330
	Bottle NM	4315	LDPE	Amber	38	27	153	335
	Bottle NM	4315	HDPE	Natural	38	27	153	330
	Bottle WM with handle	4500	PP	Natural	100	86.7	163	293
	Bottle WM with handle	4500	HDPE	Natural	100	86.7	163	293
	Heavy Duty Vacuum Bottle	4100	PP	Natural	83	63.5	154	336
5 litre	Aspirator Bottle W/Stopcock	5400	PP	Natural	54	37	169	345
10 litre	Aspirator Bottle W/stopcock	10500	PP	Natural	54	43	196	454
	Carboy NM ROUND	11500	PP	Natural	83	64	250	376
	Carboy NM ROUND W/stopcock	11100	PP	Natural	83	64	250	376
	Carboy WM ROUND	11400	PP	Natural	100	85	250	331
20 litre	Carboy NM ROUND	23100	PP	Natural	83	64	288	520
	Carboy NM ROUND W/stopcock	23000	PP	Natural	83	64	288	520
	Carboy WM ROUND	23000	PP	Natural	100	85	285	472

Resin Quick Reference Chart

	Polypropylene (PP)	Polypropylene Copolymer (PPCO)	Low Density Polypropylene (LDPE)	High Density Polypropylene (HDPE)
High Temparature	135°C	121°C	80°C	120°C
Low Temparature	0°C	-40°C	-100°C	-100°C
Autoclavable	Y	Y	N	N
Microwavable	Y	Marginal	Y	N
Dry Heat (Oven)	N	N	N	N
Freeze	N	Y	Y	Y
Flexibility	Rigid	Moderate	Excellent	Rigid
Clarity	Translucent	Translucent	Translucent	Translucent
Recycling Symbol	5 PP	5 PP	4 LDPE	2 HDPE

Autoclaving Conditions

121°C

20 min

PSI 15 psi

CAUTION

Before autoclaving set the closure on top of the container without engaging the threads. During decompression phase of the autoclaving cycle, the pressure within the vessel must be allowed to equalize. Any material stacked or placed over the closure has a potential to cause a vacuum to form, resulting in collapse.

Physical Properties of PlastX Labware

Resin	Max Use Temp °C	Brittleness Temp °C	Transparency	Sterilization					Specific Gravity	Flexibility	Water Absorption %
				Auto-claving	Gas	Dry Heat	Radia-tion	Disin-fectant			
HDPE	120	-100	Translucent	No	Yes	No	Yes	Yes	0.95	Rigid	<0.01
LDPE	80	-100	Translucent	No	Yes	No	Yes	Yes	0.92	Excel	<0.01
PC	135	-135	Clear	Yes +	Yes	No	No	Yes	1.20	Rigid	0.35
PMMA	50	20	Clear	No	Yes	No	Yes	Some	1.20	Rigid	0.30
PP	135	0	Translucent	Yes	Yes	No	No	Yes	0.90	Rigid	<0.02
PF	90	20	Clear	No	Yes	No	Yes	Some	1.05	Rigid	0.05
PSF	165	-10	Clear	Yes	Yes	Yes +	Yes	Yes	1,24	Rigid	0.30
PTFE	270	-200	Opaque	Yes	Yes	Yes	No	Yes	2.17	Excel	<0.01
PVDF	110	-62	Translucent	Yes	Yes	No	No	Yes	1.75	Rigid	0.05
TPX*	175	20	Clear	Yes	Yes	Yes +	No	Yes	0.83	Rigid	<0.01

Sterilization

- Autoclaving (121°C, 15 psig for 20 minutes) Clean and rinse item with distilled water before autoclaving. Cerntain Chemicals which have no appreciable effect on resins at room temperature may cause deterioration at autoclaving temperatures.
  ALWAYS COMPETELY DISENGAGE THREADS BEFORE AUTOCLAVING.
- Gas - Ethylene oxide, formaldehyde.
- Dry heat - 160°C for 120 minutes
- Disinfectant - Benzalkonium chloride, Formalin, Ethanol, etc.
- Radiation - gama irradiated at 1 Mrd = 10 KGY with unstabilized plastic.
- Sterilizing reduces mechanical strength. Do not use PC vessels for vacuum application if they have been autoclaved.

Chemical Resistance for PlastX Labware

Classes of Substances at Room Temperature	HDPE	LDPE	PC	PMMA	PP	PS	PSF	PTFE	PVDF	TPX*
Acids, Dilute or Weak	E	E	E	G	E	E	E	E	E	E
Acids, Strong and Concentrated	E	E	N	N	E	F	G	E	E	E
Alcohols, Aliphatic	E	E	G	N	E	E	G	E	E	E
Aldehydes	G	G	F	G	G	N	F	E	E	G
Bases	E	E	N	F	E	E	E	E	E	E
Esters	G	G	N	N	G	N	N	E	G	G
Hydrocarbons, Aliphatic	G	F	F	G	G	N	G	E	E	F
Hydrocarbons, Aromatic	G	F	N	N	F	N	N	E	E	F
Hydrocarbons, Halogenated	F	N	N	N	F	N	N	E	N	N
Ketones	G	G	N	N	G	N	N	E	N	F
Oxidizing Agents, Strong	F	F	N	N	F	N	G	E	G	F

Chemical Resistance Classification
E = Excellent - 30 days of constant exposure cause no damage. Plastics may even tolerate for years.
G = Good - Little or no damage after 30 days of constant exposure to the reagent.
F = Fair - Some effect after 7 days of constant exposure to the reagent like crazing, cracking, loss of strength or discoloration.
N = Not recomended - Not for continuous use. immediate damage may occur.

Resin Code

HDPE	High-density polyethylene
LDPE	Low-density polyethylene
PC	Polycarbonate
PMMA	Polymethyl methacrylate
PP	Polypropylene
PS	Polystyrene
PSF	Polysulfone
PTFE	Polytetrafluoroethylene
PVDF	Polyvinylidene fluoride
TPX	Polymethylpentene

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Leak Testing

Leakage test by Vacuum Method

Product Criterion

Bottles / Vials upto 1000 ml.

Sample Size

One full cycle shot quantity (Twice per shift) or as defined by customer sampling plan.

Test Equipment

Vacuum Desiccator.

Test Procedure

1. Fill the bottle / vial with water up to 80% of its brimful volume and secure tightly with closure by hand.
2. Invert the bottle / vial and visually check for any leakage.
3. Keep the bottle for 3 - 5 min in Vacuum Desiccator at 635 mm of Hg.
4. Release the vacuum and check for any trace of water on the threads of bottle or closure.
5. Open the closure and check for any trace of water on the threads of bottle or closure.
6. If no leakage is observed in the testing, the bottles are certified leakproof.

Leakage test by Gravity Method

Product Criterion

Bottles / Containers above 1000 ml.

Sample Size

One full cycle shot quantity (Twice per shift) or as defined by customer sampling plan.

Test Procedure

1. Fill the bottles with water up to 80% of its brimful volume and secure tightly with closure by hand.
2. Invert the bottle and visually check for any leakage.
3. Leave the bottle for 24 hours in inverted condition.
4. After 24 hours, check bottle for any leakage.
5. Open the closure and check for any trace of water on the threads of bottle or closure.
6. If no leakage is observed in the testing, the bottles are certified leakproof.

Autoclaving Instructions

PlastX recommends an autoclaving cycle is 121°C, 15 psi for 20 minutes for its Polypropylene, Polycarbonate and Polymethylpentene.

Autoclaving

Procedure

a) Autoclave testing to be carried out in wet method for Polypropylene bottles and Polycarbonate flasks having 75% water filled.
b) Autoclave testing to be carried out in dry method for Polypropylene Micro Tips, Centrifuge Tubes, Funnel, etc.
c) Autoclave testing to be carried out in dry method for Polymethylpentene Lab Wares.

Test Parameters

1. Temperature = 121°C.
2. Pressure = 15 psi.
3. Time = 20 minutes.

Important Notes

- Before autoclaving, just set closure on top of the container without engaging the threads (if this is not done, pressure differentials may cause containers to collapse during autoclaving).
- For best results, use a slow exhaust cycle.
- Clean and rinse item with distilled water before autoclaving.
- Sterilizing reduces mechanical strength. Do not use PC vessels for vacuum application during autoclaving.
- Certain chemicals which have no appreciable effect on resins at room temperature may cause deterioration at autoclavable temperature.

Precautions while autoclaving Polycarbonate

- Polycarbonate must be thoroughly rinsed before autoclaving because detergent residues cause crazing and spotting.
- The cycle should be limited to 20 minutes at 121°C since PC shows loss of mechanical strength after repeated autoclaving.

Practices to avoid during Autoclaving

1. **DO NOT** stack the bottles, jars and carboys.
2. **DO NOT** place the product in an autoclaving basket with other objects placed on top.
3. **DO NOT** tighten the closure – removing closure is safer.
4. **DO NOT** place aluminum foil, gauze, cotton, tape or sterile wraps over the opening.

The guidelines are for empty containers only. PlastX does not provide any validation / autoclaving manual/ instructions/information on autoclaving with liquid inside any of our products. The consumer must perform all validation work regarding the procedure.

We assist with the selection of resins and list out the recommended containers for your applications.

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